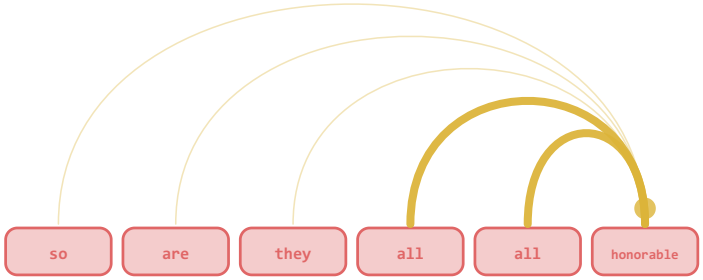
UNIT 01 · CHAPTER 8

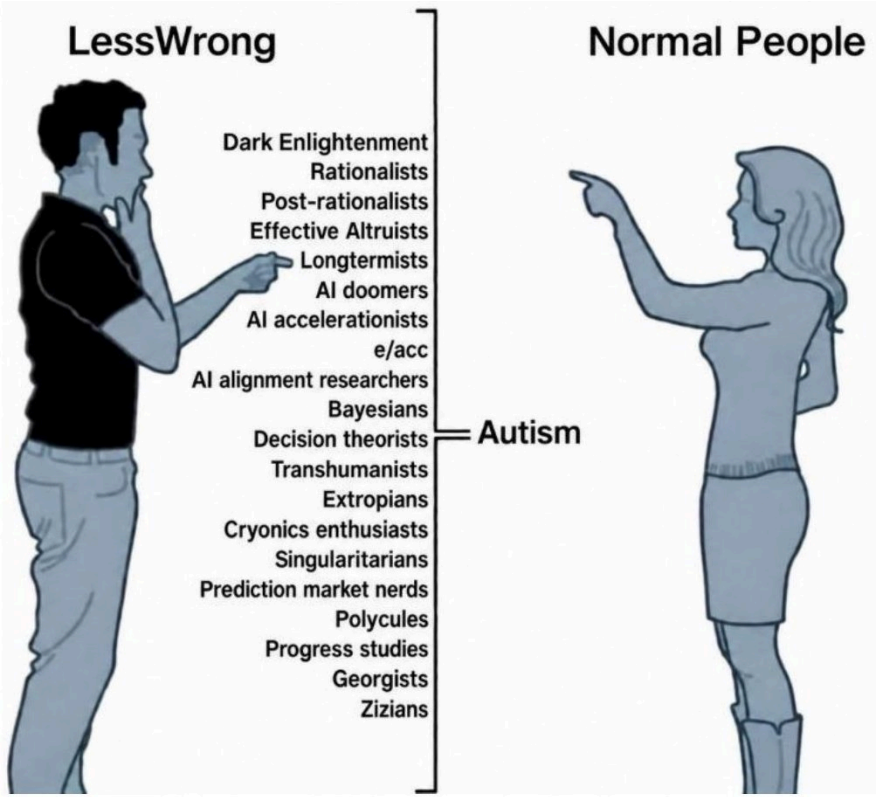
Self-Attention

CSE 10124 · Building ChatGPT

William Theisen · University of Notre Dame

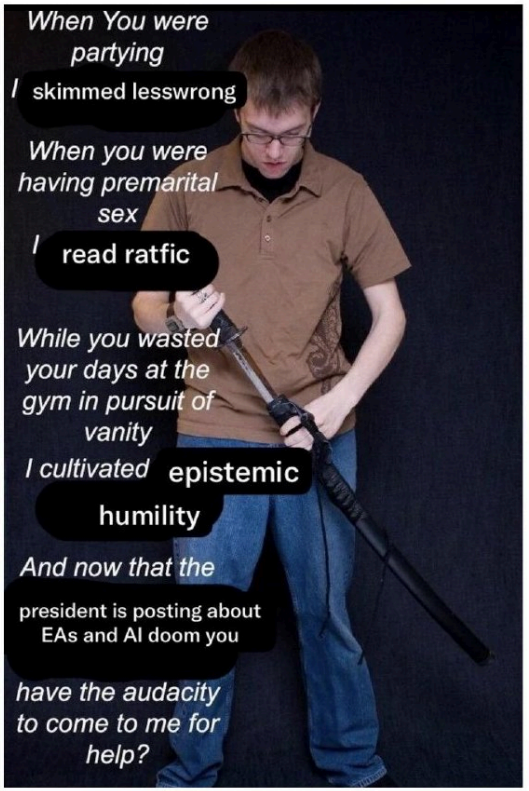


In the News


LessWrong
Normal People
Dark Enlightenment
Rationalists
Post-rationalists
Effective Altruists
Longtermists
AI doomers
AI accelerationists
e/acc
AI alignment researchers
Bayesians
Decision theorists
Transhumanists
Extropians
Cryonics enthusiasts
Singularitarians
Prediction market nerds
Polycules
Progress studies
Georgists
Zizians
Autism

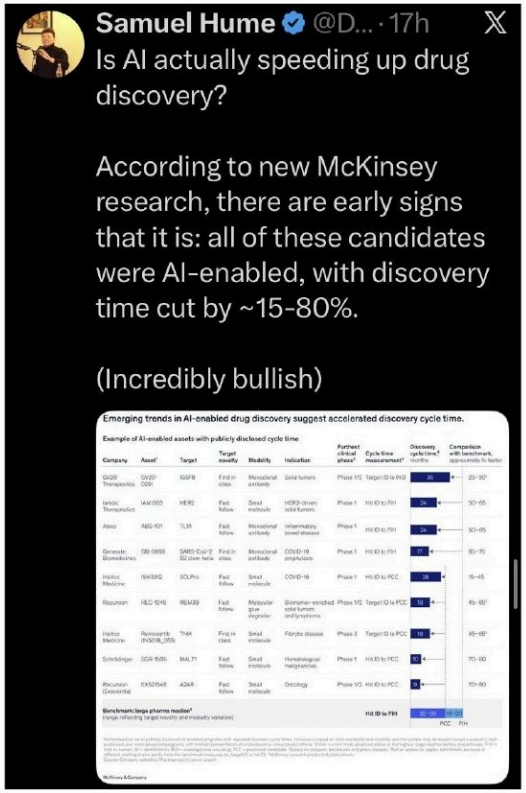
In the News


nine @atlantic... · 13h
So much San Franciscan culture is computer nerds refusing to learn about anything before 1990 and tediously relitigating entire intellectual histories except in their annoying reddit dialect and cadence.
They call this “working from first principles”
corsaren @corsar... · 15h
effective altruist gf: “would you still love me if i was 10 billion worms?”
35 66 1.5K 27K

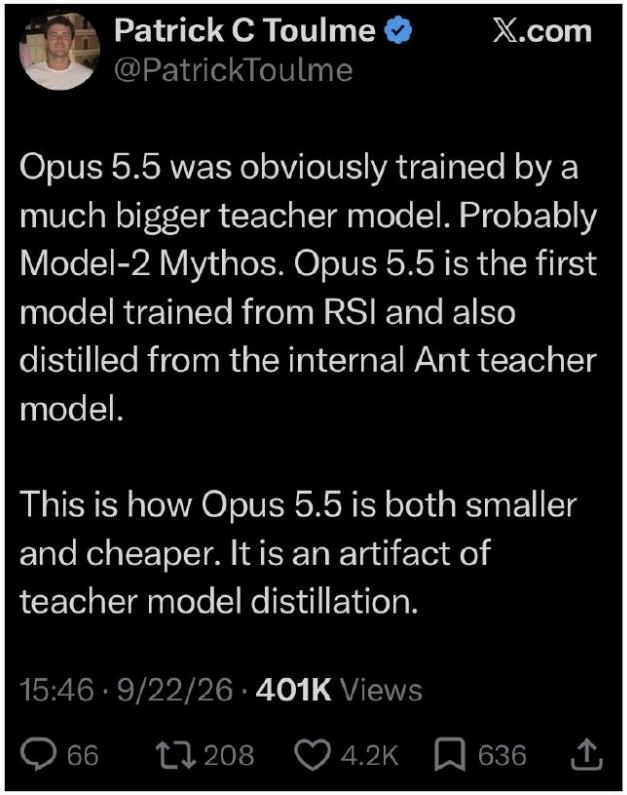

When You were partying
I skimmed lesswrong
When you were having premarital sex
I read ratfic
While you wasted your days at the gym in pursuit of vanity
I cultivated epistemic humility
And now that the president is posting about EAs and AI doom you
have the audacity to come to me for help?

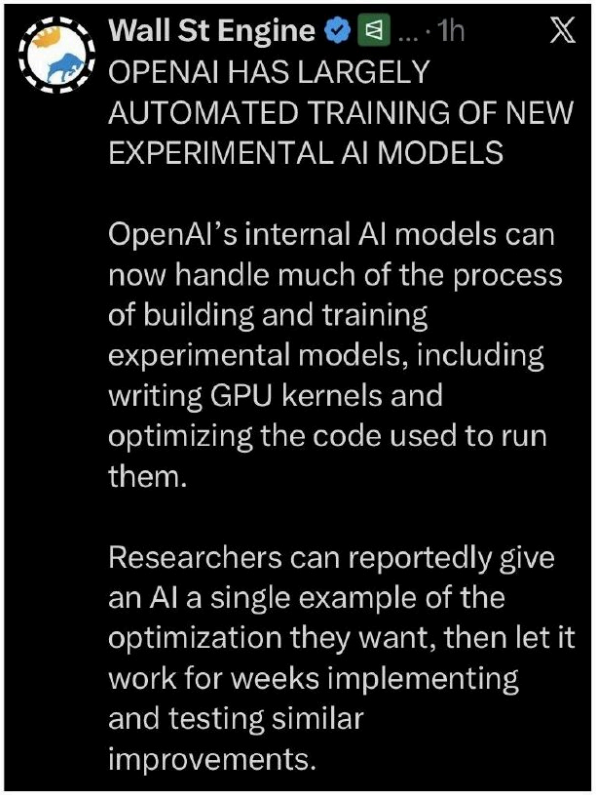
In the News


Christian Elton @... · 17h
The gap between major model releases keeps shrinking.
2023: once every 73 days
2026 (so far): once every 18 days
Apr 2024
Major releases so far
7
Days since last OpenAI release
160
Claude 3
Claude 2.1
OpenAI
Anthropic
Days since last Anthropic release
41
Claude and 3 others
90
455
4K
278K

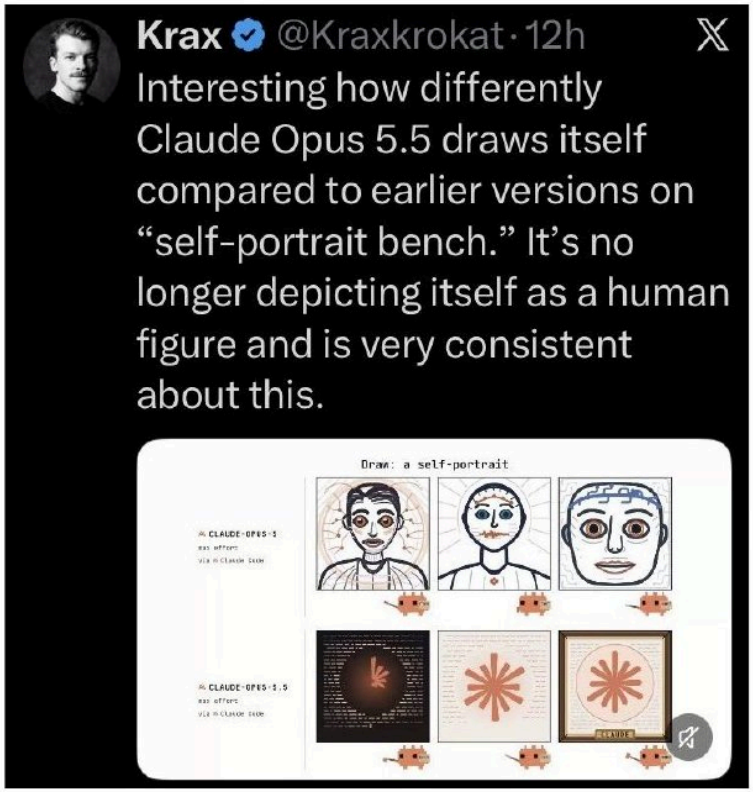

Samuel Hume @D... · 17h
Is AI actually speeding up drug discovery?
According to new McKinsey research, there are early signs that it is: all of these candidates were AI-enabled, with discovery time cut by ~15-80%.
(Incredibly bullish)
Emerging trends in AI-enabled drug discovery suggest accelerated discovery cycle time.
Example of AI-enabled assets with publicly disclosed cycle time
Company
Asset
Target
Target novelty
Modality
Indication
Furthest clinical phase
Cycle time measurement
Discovery cycle time, months
Comparison with benchmark, approximate % faster
Benchmark: large pharma median
Hit ID to FIH
PCC
FIH

In the News


Patrick C Toulme
@PatrickToulme
X.com
Opus 5.5 was obviously trained by a much bigger teacher model. Probably Model-2 Mythos. Opus 5.5 is the first model trained from RSI and also distilled from the internal Ant teacher model.
This is how Opus 5.5 is both smaller and cheaper. It is an artifact of teacher model distillation.
15:46 · 9/22/26 · 401K Views
66
208
4.2K
636


Wall St Engine
... · 1h
OPENAI HAS LARGELY AUTOMATED TRAINING OF NEW EXPERIMENTAL AI MODELS
OpenAI's internal AI models can now handle much of the process of building and training experimental models, including writing GPU kernels and optimizing the code used to run them.
Researchers can reportedly give an AI a single example of the optimization they want, then let it work for weeks implementing and testing similar improvements.

In the News

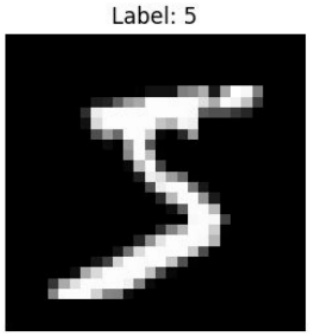

Krax @Kraxkrokat · 12h
Interesting how differently Claude Opus 5.5 draws itself compared to earlier versions on “self-portrait bench.” It’s no longer depicting itself as a human figure and is very consistent about this.
Draw: a self-portrait

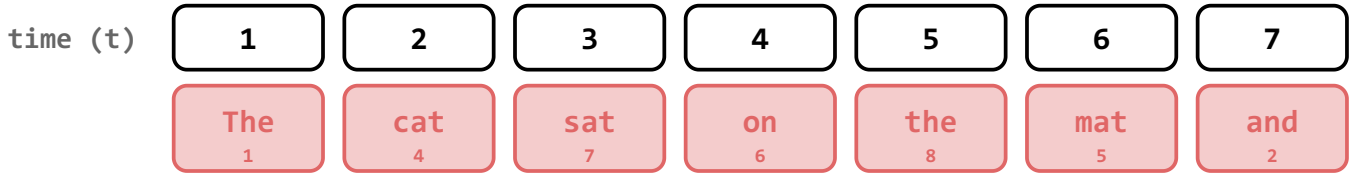

bashu, thanks @b... · 7h
Being a dev today using frontier AI feels like I'm playing Watson to Sherlock Holmes. I just stand back and go "my word holmes how did you deduce that" and occasionally stop him from getting shot
13 254 3.6K 53K


Disclose.tv @dis... · 37m
JUST IN - Trump renames AI to SI: "Super Intelligence."
126 154 761 79K

Reading vs. Seeing

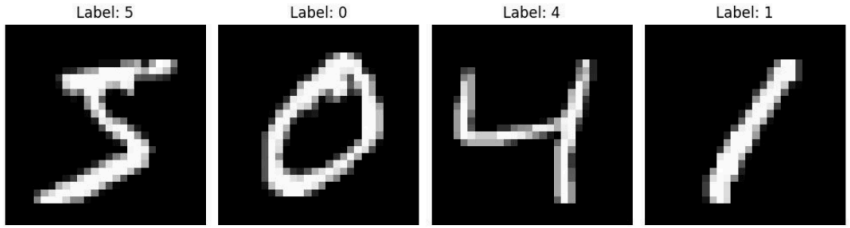
When we **see** something, we generally take it in all at once.





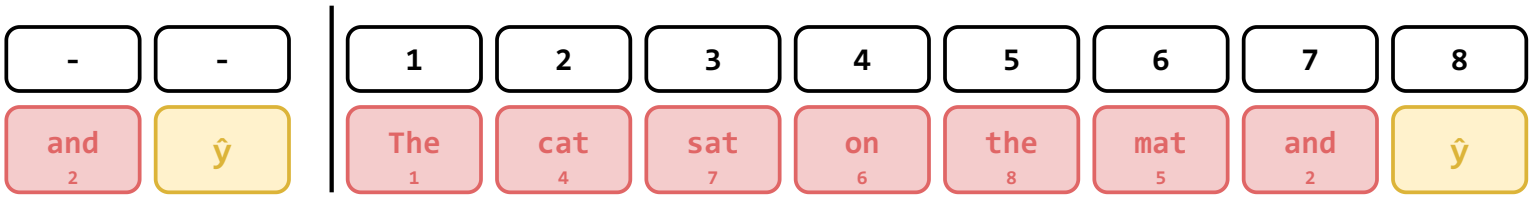
However, when we **read**, we typically read in a sequence, one word at a time. We could say that reading has a dimension of `time (t)`.

Reading vs. Seeing

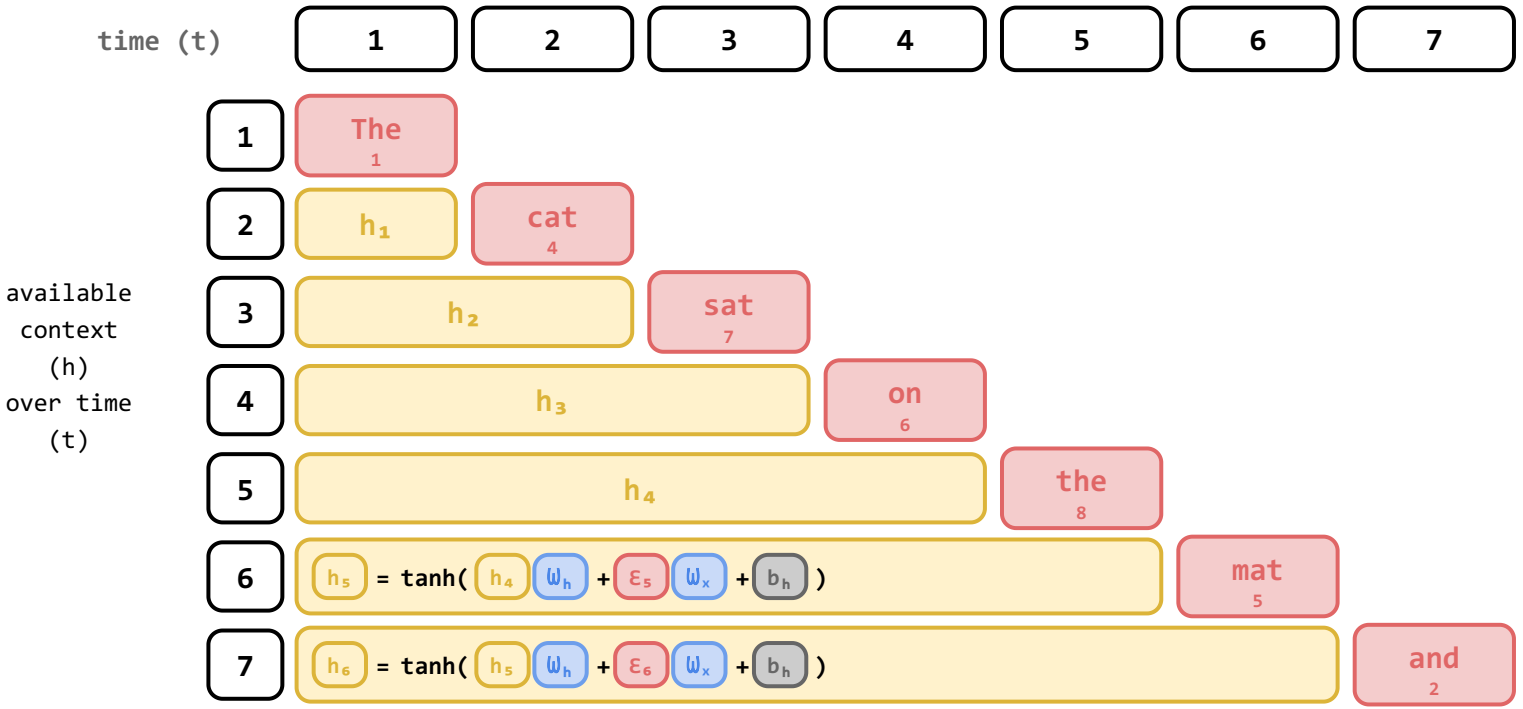


When we're looking at images, it doesn't really matter what image we saw previously.

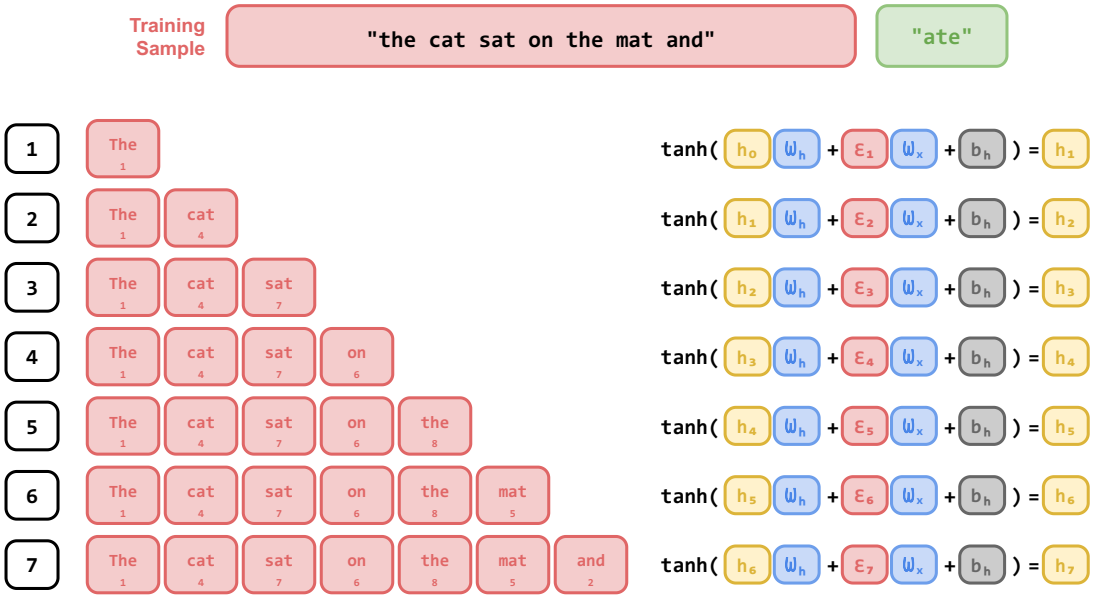
However, as we read across `time (t)`, what we've seen previously matters a LOT! It **provides context to the token we're currently seeing**.



Time and Context in RNNs



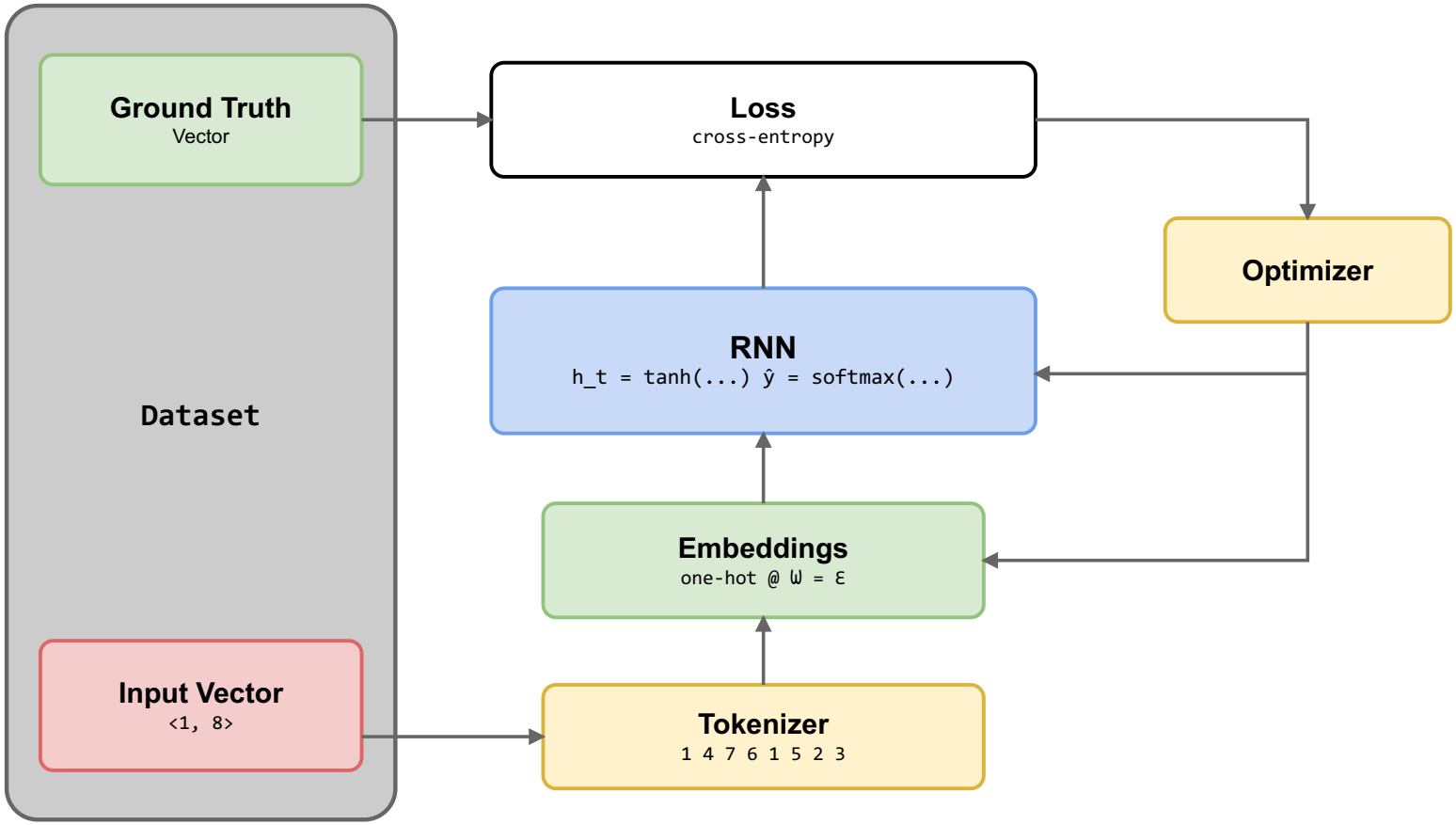
RNNs: Producing output at time (t)



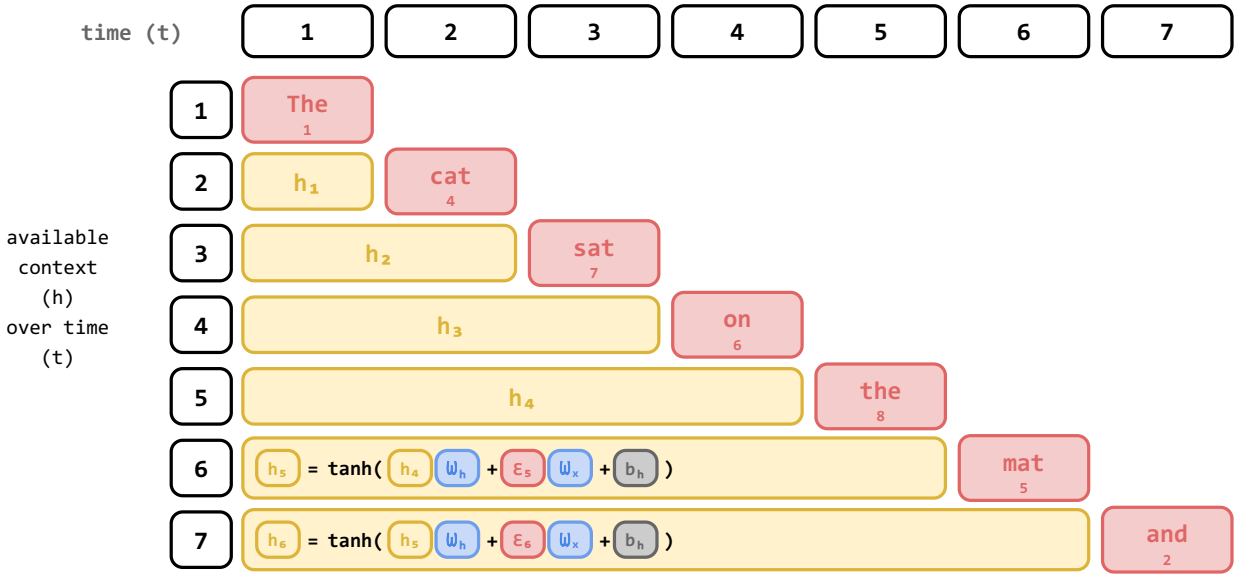
$$\hat{y} = \mathrm{softmax}(h_7 W_y + b_y)$$

Token ID	0	1	2	3	4	5	6	7	8
Token	"."	"The"	" and"	" ate"	" cat"	" mat"	" on"	" sat"	" the"
$\hat{y}$	0.05	0.07	0.03	0.08	0.12	0.10	0.10	**0.38**	0.07

Training an RNN



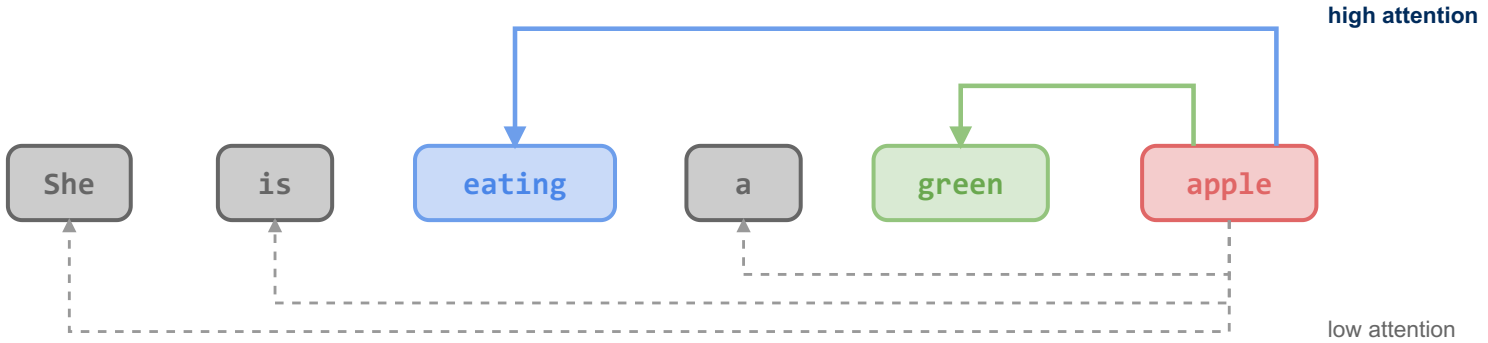
Weakness of RNNs



If I want to measure how "cat" should affect "and", I have to go back through 5 "context blocks" (h_2 to h_6). Since these are "compressions" we lose a lot of meaning/context from things earlier in the sequence. RNNs struggle with "long-term dependencies".

Attention

Some parts of our **context** are **more important than others**!



We can calculate how important embeddings are to each other using **self-attention** and then we can update each embedding with context!

Self Attention

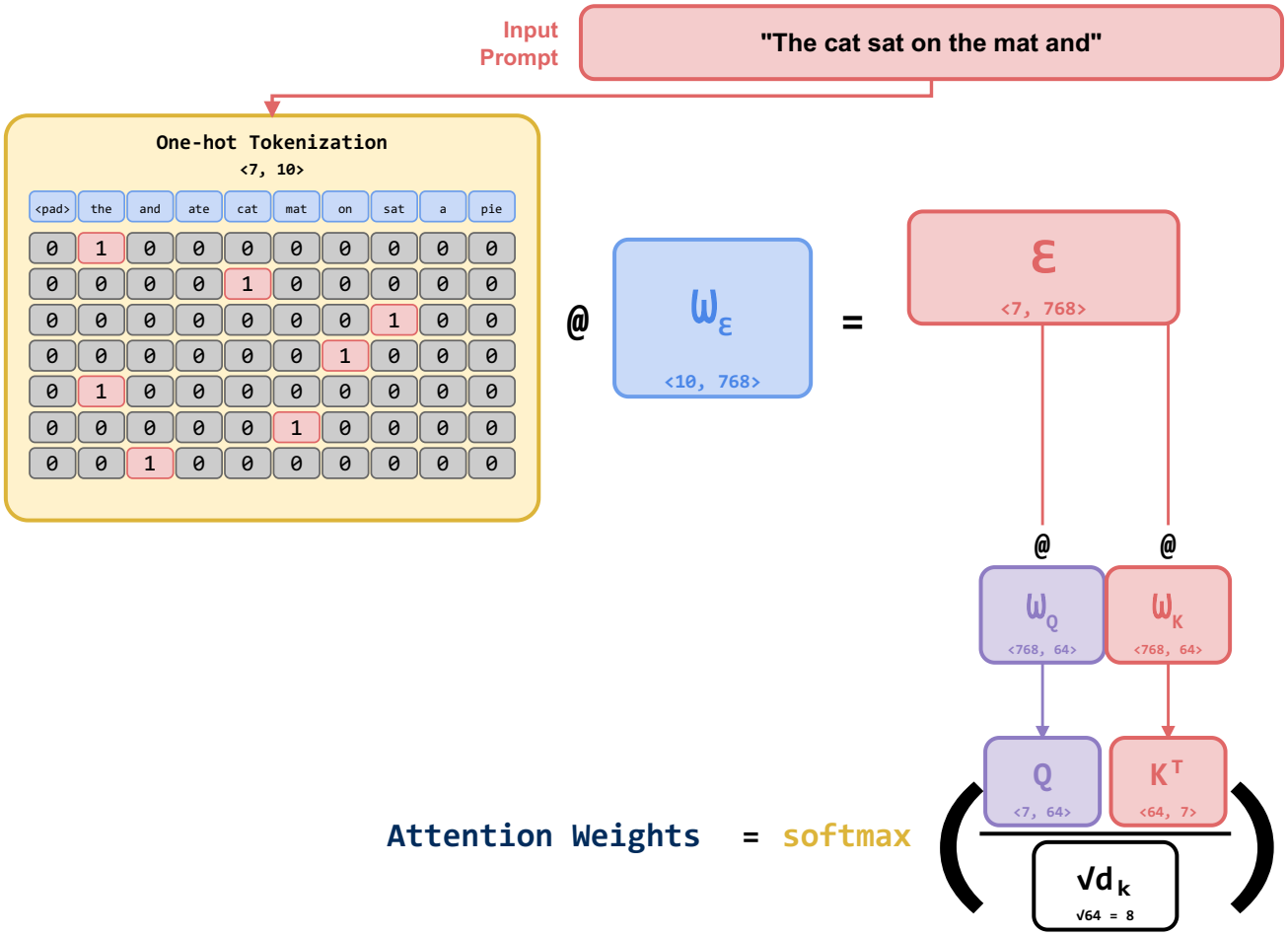
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q \; K^T}{\sqrt{d_k}}\right) V$$

Q: <7, 64> · K^T: <64, 7> · $\sqrt{d_k}$: $\sqrt{64} = 8$ · V: <7, 768>

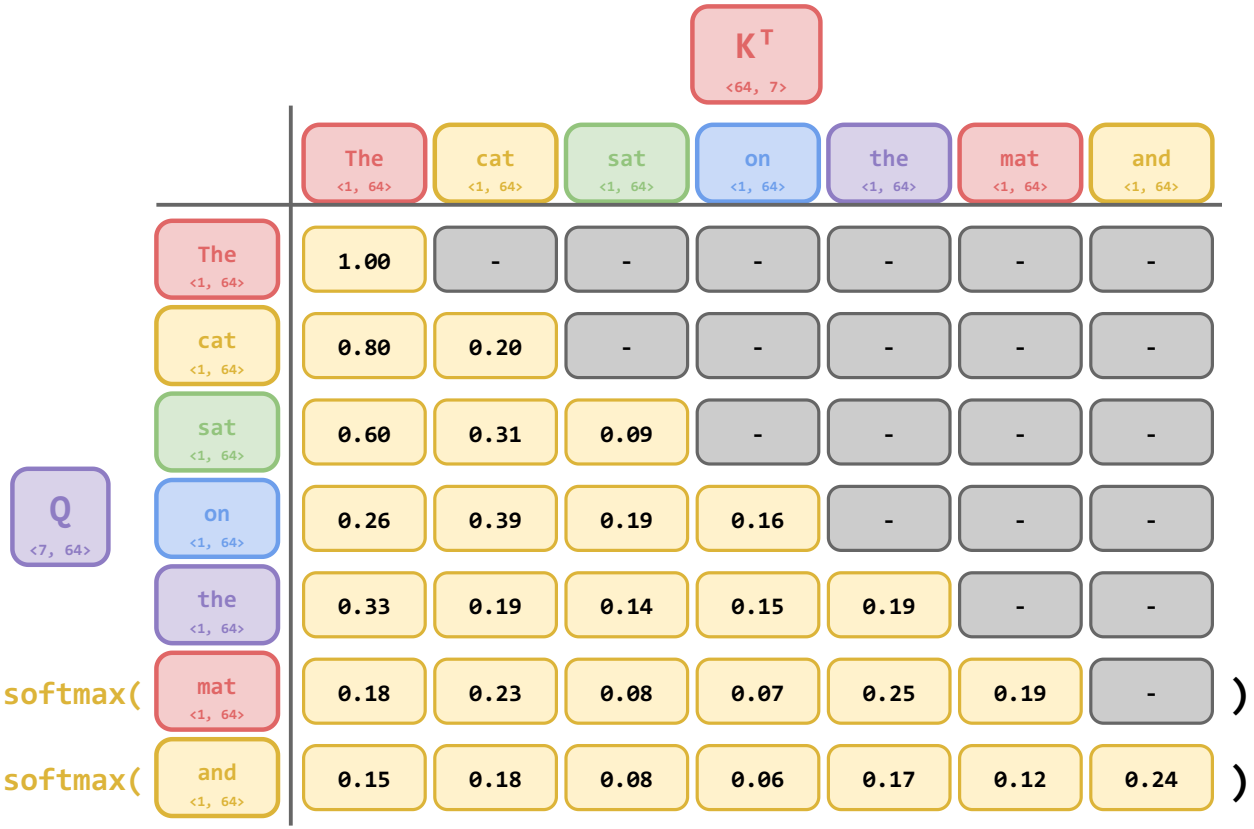
They mentally model this as a "**dictionary lookup**"

- **Q – Query**: What each embedding is "looking" for
- **K – Key**: What each embedding "has"
- **V – Value**: The "context" that each embedding actually gives

Calculating attention weights



Calculating attention



Each row is a measure of how much attention that token should pay to each other token (columns) out of 1, almost like a "percentage" of your total attention.

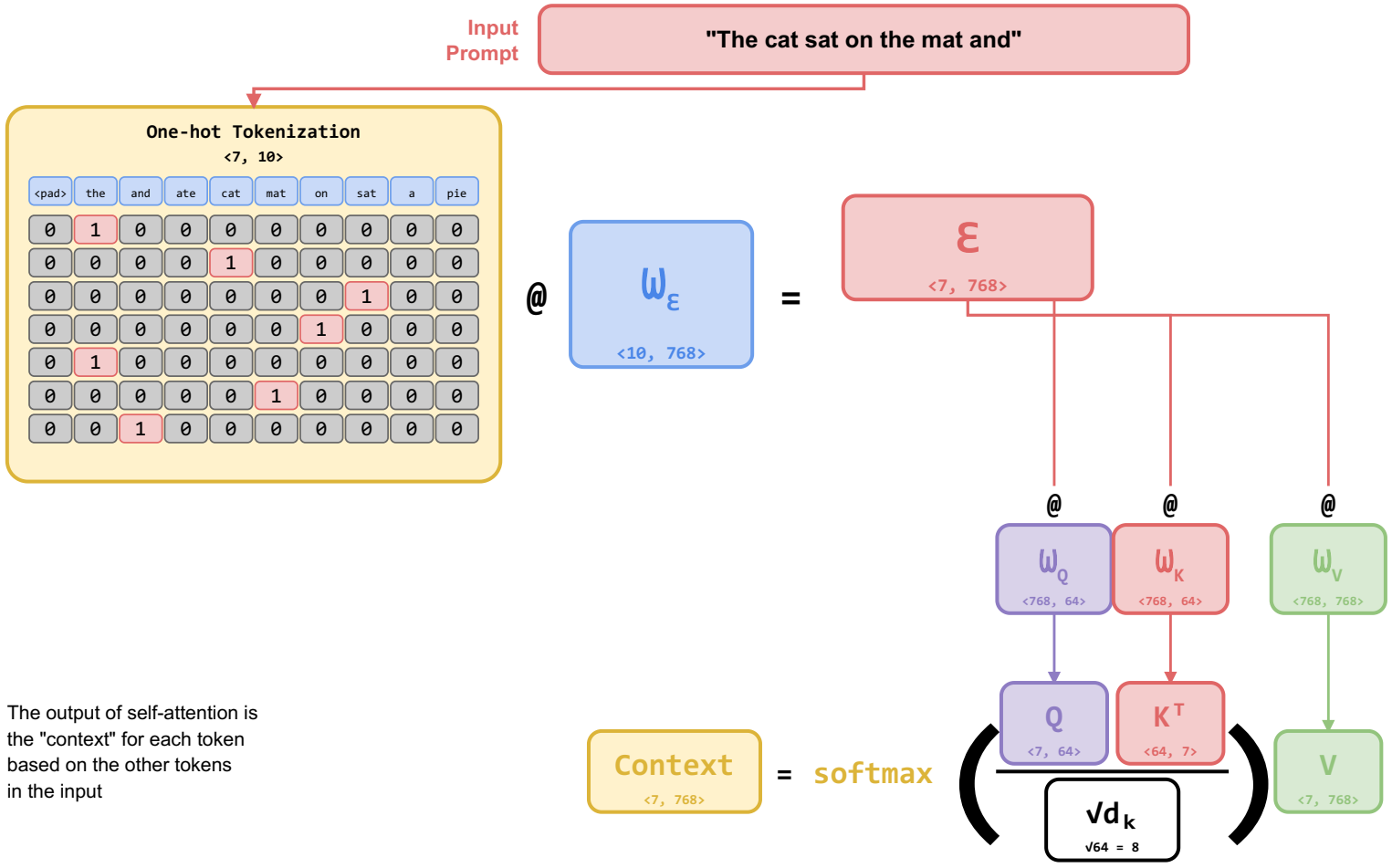
Attention Equation



To continue with the **dictionary lookup** analogy (which I've always found a bit confusing):

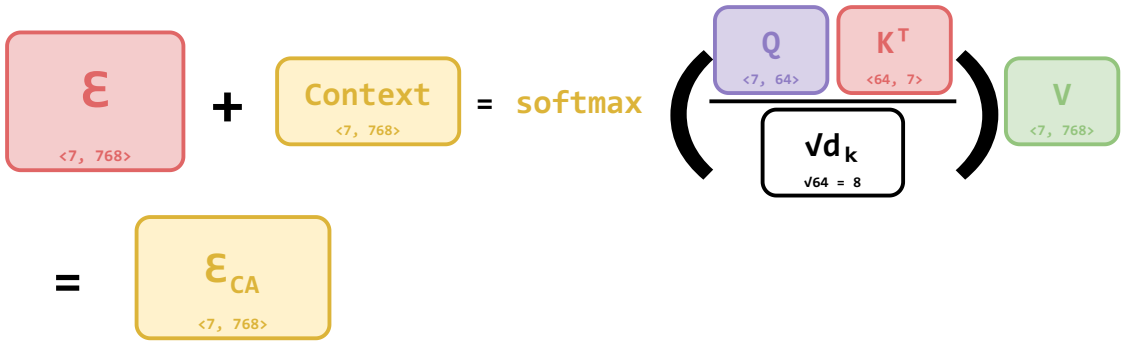
- $Q\ K^T$: How much each embedding matters to each other embedding (called attention scores)
- $\sqrt{d_k}$: Scaling factor (technically not required)
- **softmax**: Convert the above attention scores to attention weights summing to `1`
- V: The actual context from each embedding we'll add to the original embeddings

Calculating attention



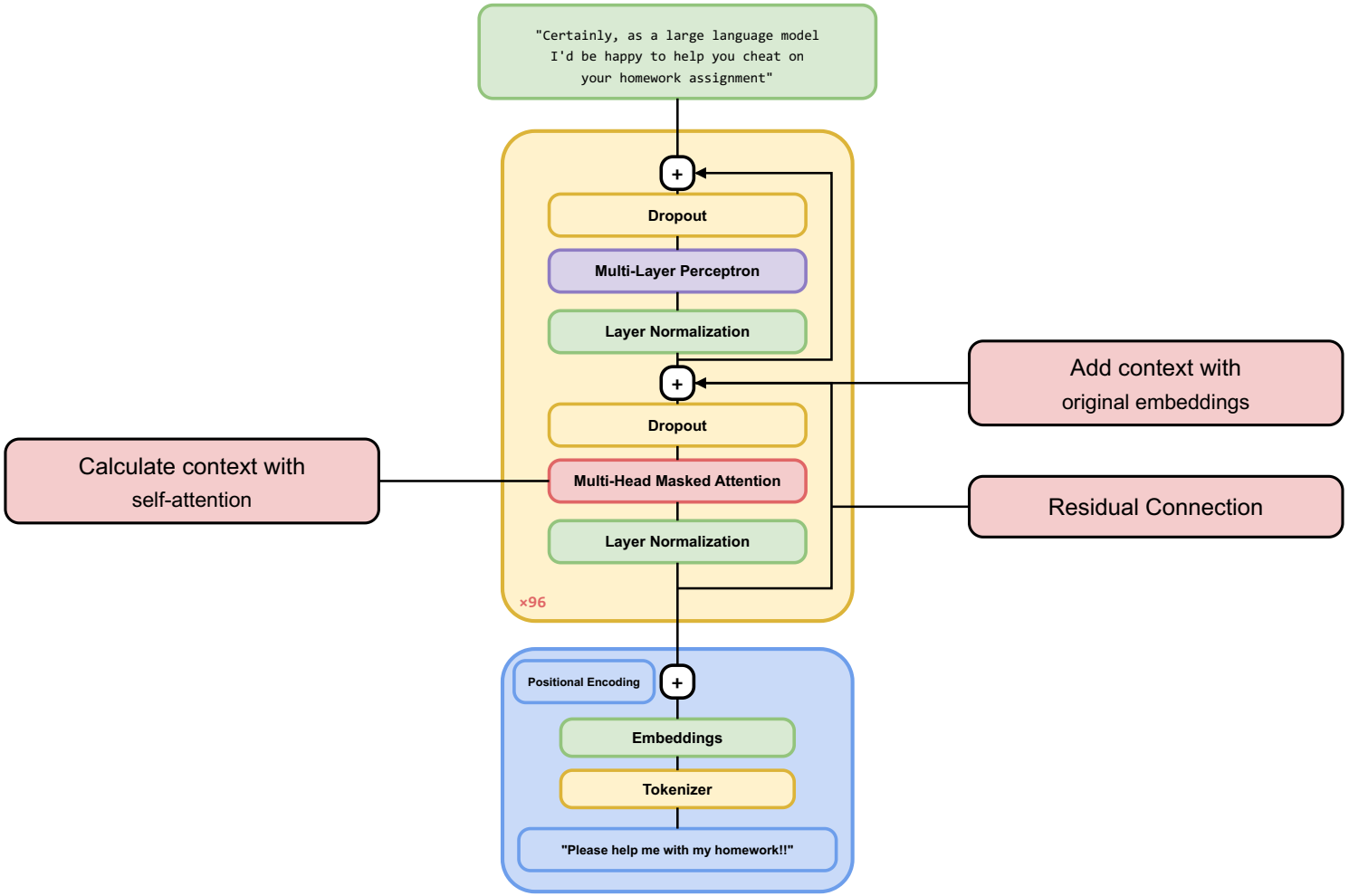
The output of self-attention is the "context" for each token based on the other tokens in the input

Residual Connections

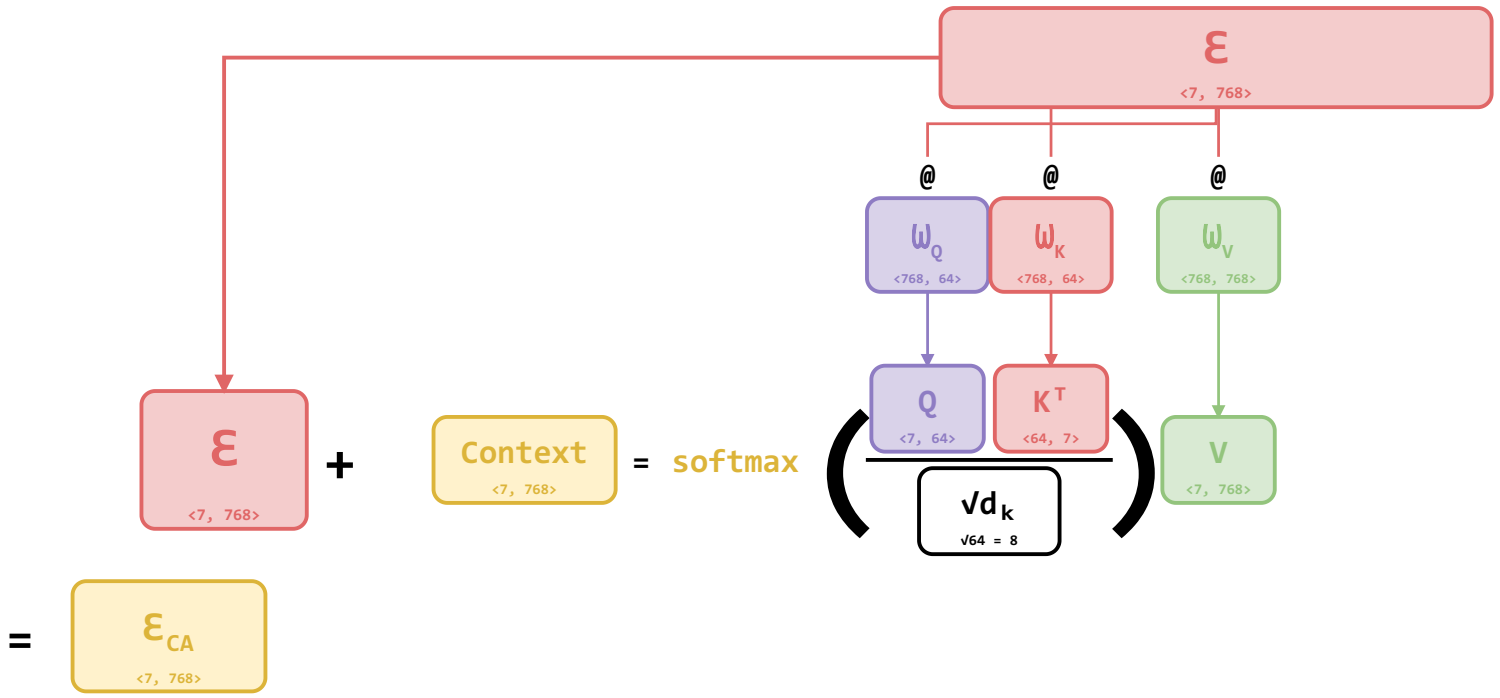


We don't want our **embeddings** to "forget" what they were originally representing, so we add them back in with the **output of self-attention**. This is called a "**residual connection**".

CSE 10124: Building ChatGPT



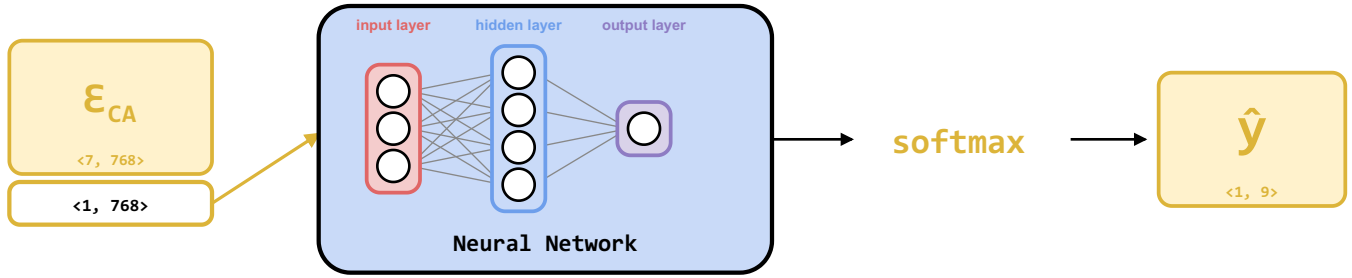
Putting it all together



ε_{CA} = Context-aware embeddings. All of our original embeddings have been "updated" to include the context of all the other tokens in the input.

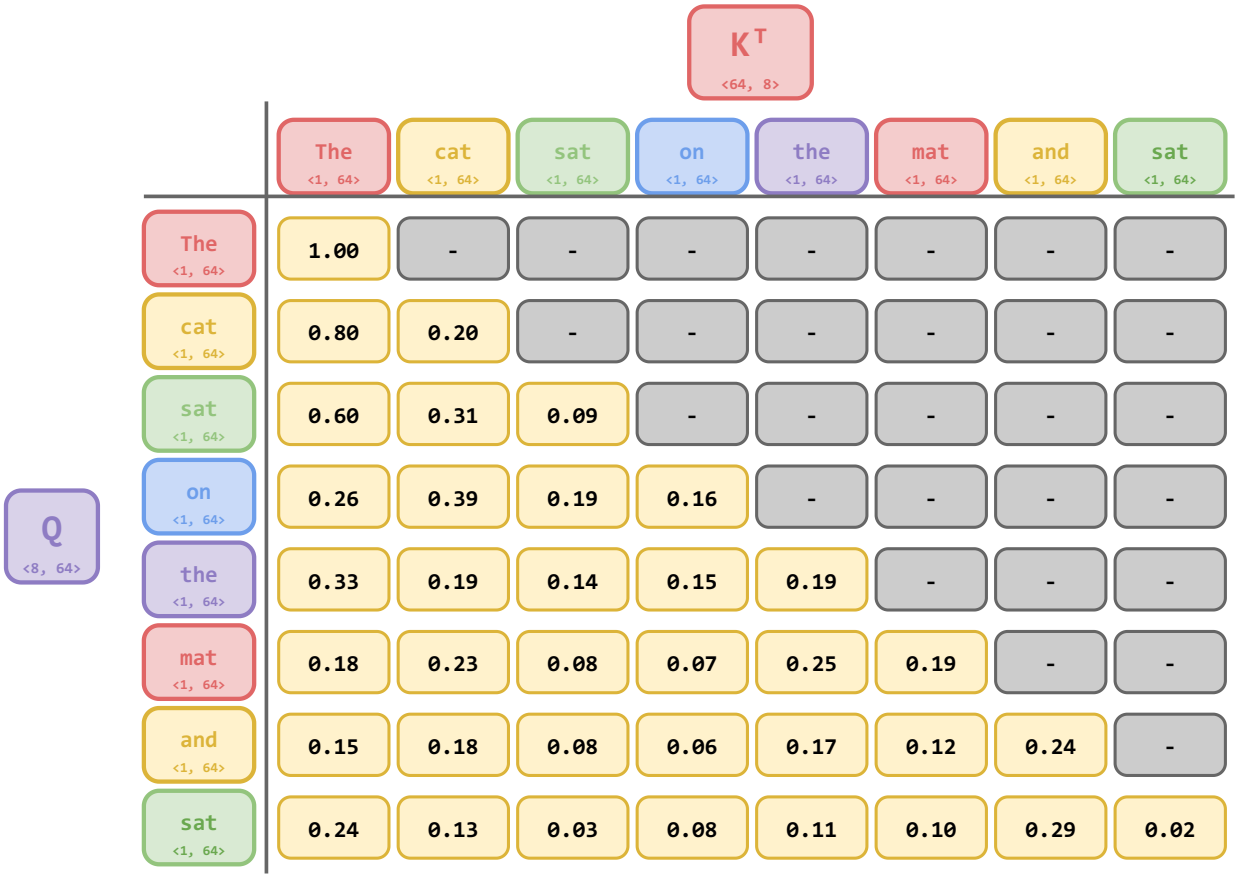
Producing output from a transformer

With our **context-aware embeddings**, we can then pass those into a simple **feed-forward neural network** to produce an **output prediction** (similar to producing an output with an RNN). If we only wanted to predict the next word, we only need the last context-aware embedding (output is size 9 because we have 9 words in our vocab).



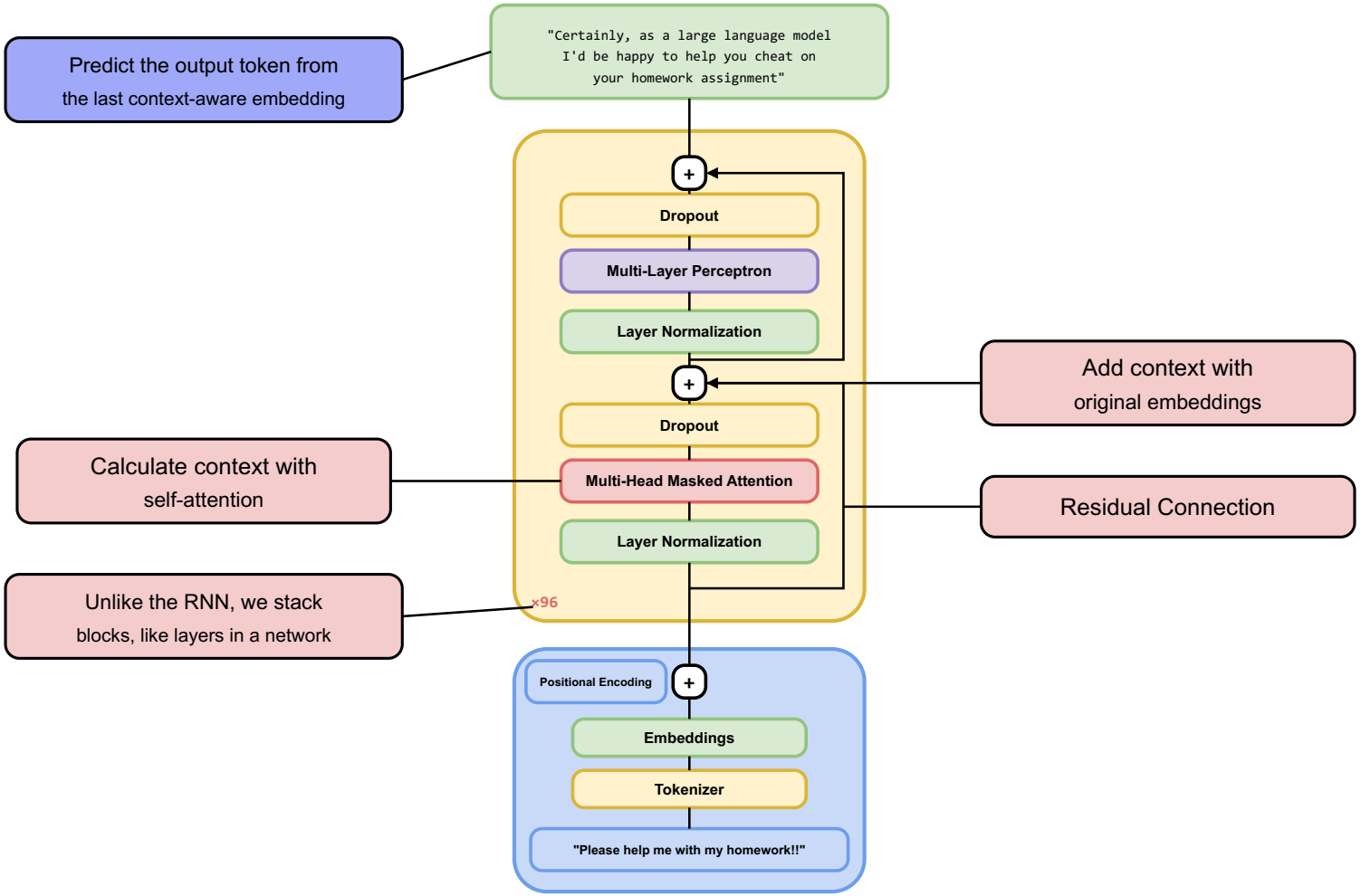
Token ID	0	1	2	3	4	5	6	7	8
Token	"."	"The"	" and"	" ate"	" cat"	" mat"	" on"	" sat"	" the"
$\hat{y}$	0.05	0.07	0.03	0.08	0.12	0.10	0.10	**0.38**	0.07

Auto-Regressive Text Generation

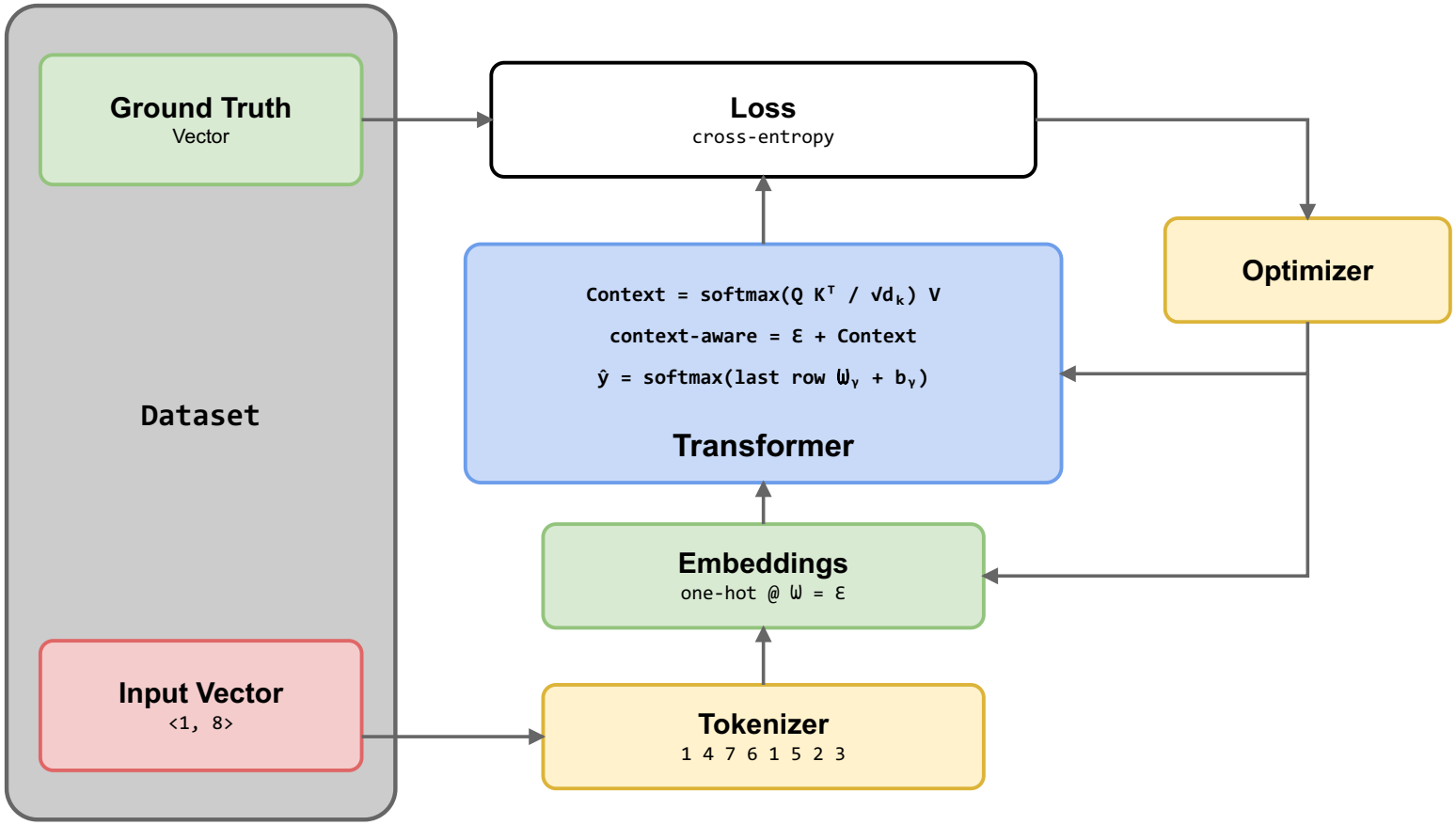


We can append the predicted token to our sequence, re-calculate attention, and then use that to predict what the next token should be, etc. etc.

CSE 10124: Building ChatGPT



Training a Transformer



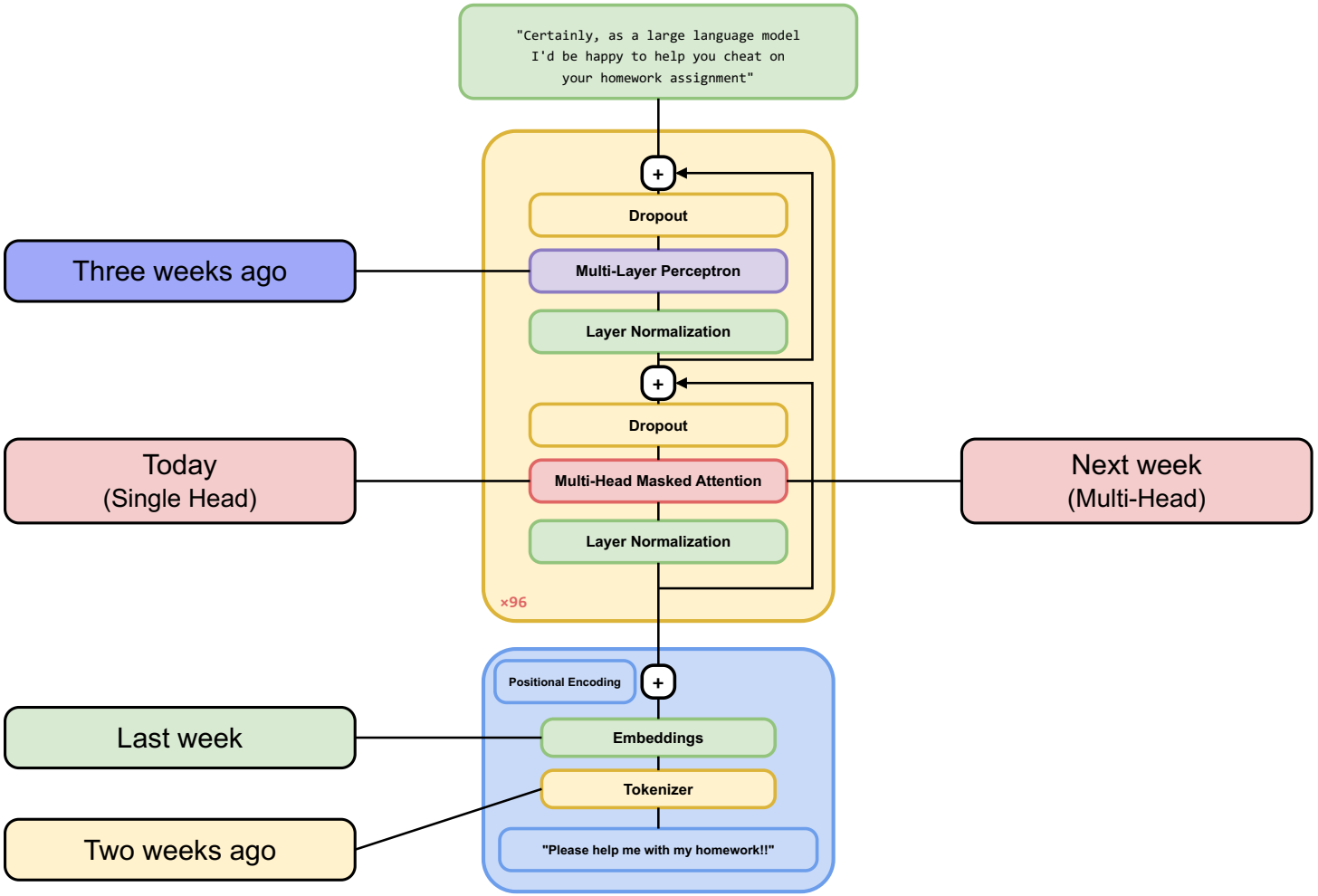
Transformers and the Brain

Schrimpf et al. (*PNAS*, 2021) compared 43 language models against people's brain activity (fMRI and electrodes) on the same sentences.

- Transformers matched best, predicting **nearly 100% of the explainable variance** in the brain's responses.
- How well a model matched the brain tracked how good it was at next-word prediction, not at other language tasks.

Careful: this is a match in *what* gets predicted, not evidence that neurons compute $\mathrm{softmax}(Q\,K^{T} / \sqrt{d_k})\,V$.

CSE 10124: Building ChatGPT



In the News

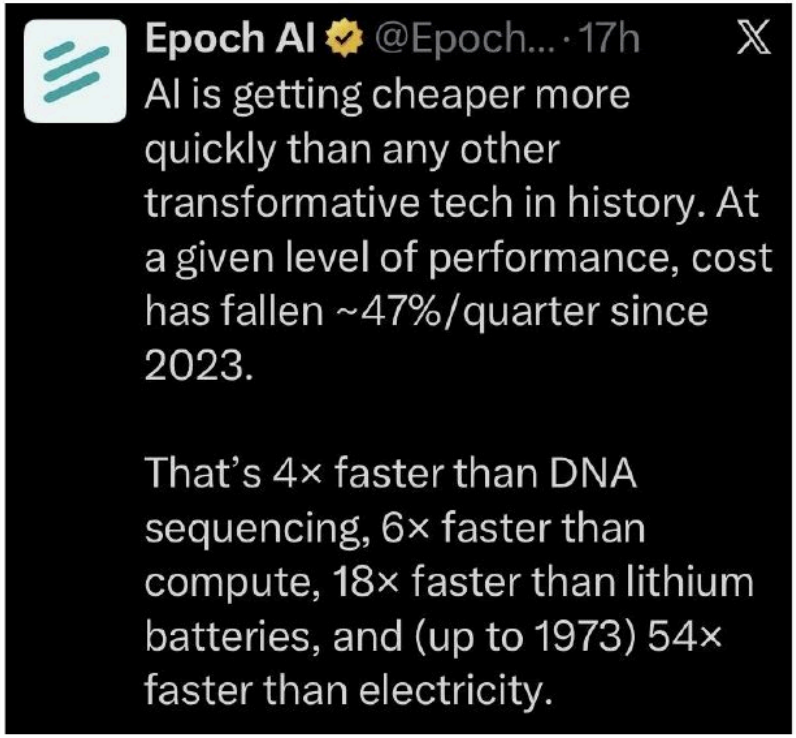

Shikhar @xikhar · 17h
Dario wrote a 3800 words article on pacing the frontier only to drop a new SOTA model 11 days later.
Safe to say, no one is actually doing it.
Pace the frontier
But not for me
48 147 3.1K 65K


Rayan Krishnan · 10h
We've updated our timeline for full RSI to July 2027 instead of August after our eval of Opus 5.5.
It's telling that the biggest advocate for pacing is still very much racing ahead. Even the most rational can fall prey to this multi polar trap. The only solution is coordination.

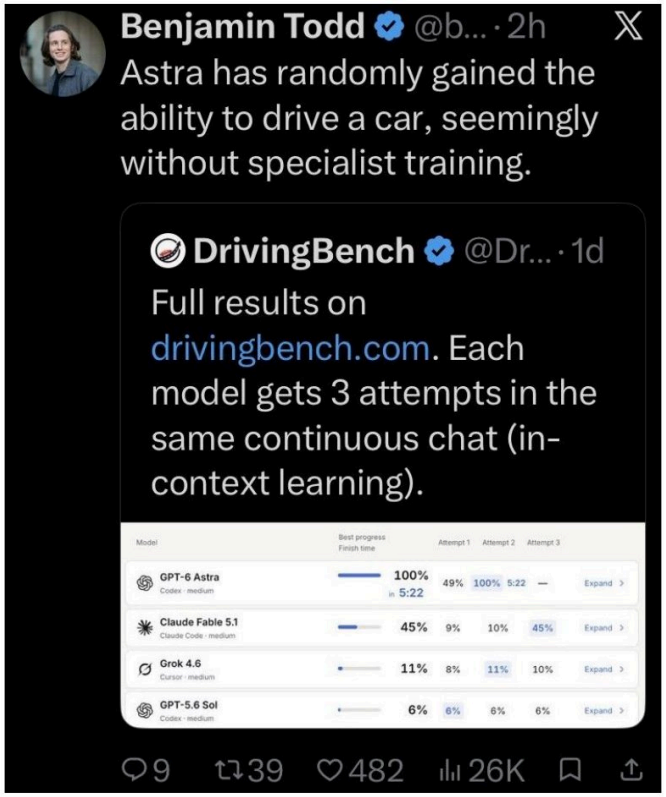

AI Notkilleveryon... · 6h
UPDATE: AIs have achieved the highest *possible* score on the Mensa Norway IQ test - 151
3 years ago: cognitively impaired human (64 IQ)
2 years ago: average human
1 year ago: genius human
Today: literally off the charts
Next year?
AI IQ Test Results

In the News


Lisa Abramowicz
@lisaabramowicz1
𝕏.com
Financing the AI buildout will total $10.3 trillion from 2025-2032, or an average of 3.63% of US GDP each year: study. That "would be larger relative to the economy than the major US canal, railroad, electrification, highway, & telecom investment booms"


Epoch AI @Epoch... · 17h
𝕏
AI is getting cheaper more quickly than any other transformative tech in history. At a given level of performance, cost has fallen ~47%/quarter since 2023.
That's 4× faster than DNA sequencing, 6× faster than compute, 18× faster than lithium batteries, and (up to 1973) 54× faster than electricity.

In the News


Benjamin Todd @b... · 2h
Astra has randomly gained the ability to drive a car, seemingly without specialist training.
DrivingBench @Dr... · 1d
Full results on drivingbench.com. Each model gets 3 attempts in the same continuous chat (in-context learning).
Model
Best progress
Finish time
Attempt 1
Attempt 2
Attempt 3
GPT-6 Astra
Codex · medium
100%
in 5:22
49%
100% 5:22
—
Expand
Claude Fable 5.1
Claude Code · medium
45%
9%
10%
45%
Expand
Grok 4.6
Cursor · medium
11%
8%
11%
10%
Expand
GPT-5.6 Sol
Codex · medium
6%
6%
6%
6%
Expand
9
39
482
26K

Self Attention Analogy



Another mental model for self-attention might be **"using tinder"**, where everyone on the app is simultaneously looking for someone and being looked at.

- **Q – Query**: What each person is looking for in a person
  - "I'm looking for a 10/10 terrachad ASU frat leader"
- **K – Key**: the first photo each person has on their profile
  - "I'm a rocket"
- **V – Value**: The rest of the profile

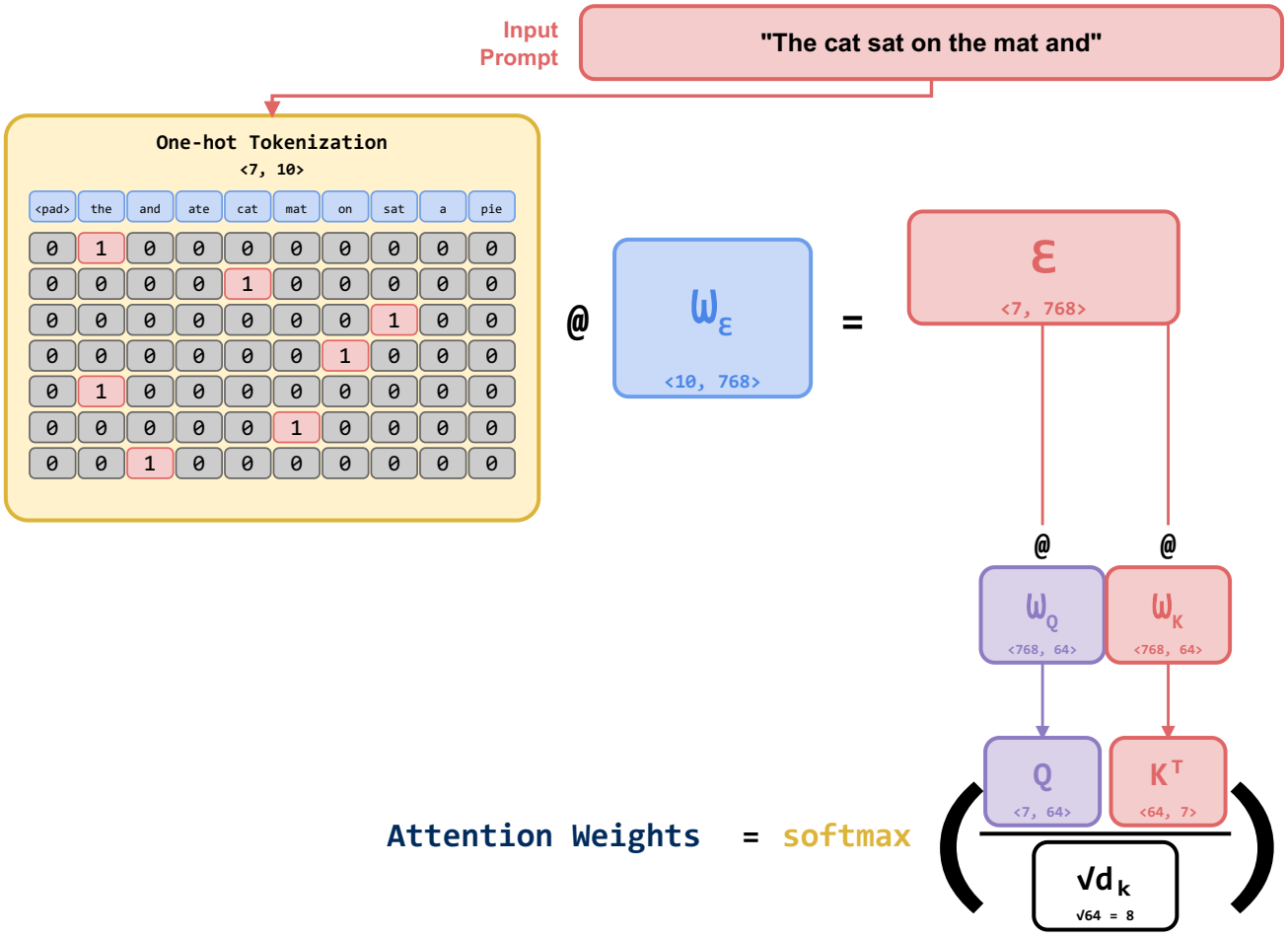
Self Attention Analogy

- **Q – Query**: What each person is looking for in a person
  - "I'm looking for a 10/10 terrachad ASU frat leader"
- **K – Key**: the first photo each person has on their profile
  - "I'm a baddie"
- **V – Value**: The rest of the profile

As you swipe through profiles, you glance at the **key** of everyone (their first photo) and calculate how much they match your **query** (ASU frat leader) while everyone else is also doing the same.

Based on this, you calculate how much **attention** you should pay to all the **values** (actual profiles) of people you see.

Calculating attention weights



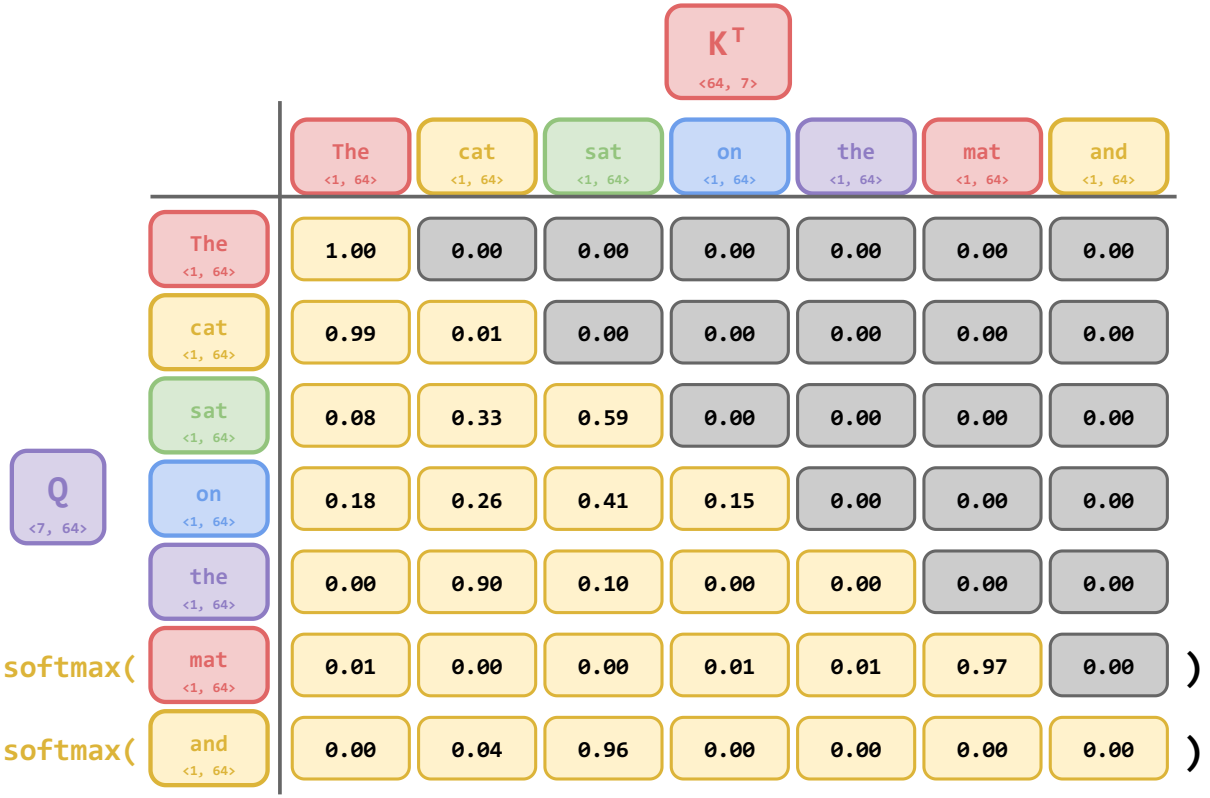
Calculating attention

Attention scores = $Q\ K^T\ /\ \sqrt{d_k}$



Calculating attention

Attention Weights = **softmax**($Q K^T / \sqrt{d_k}$)



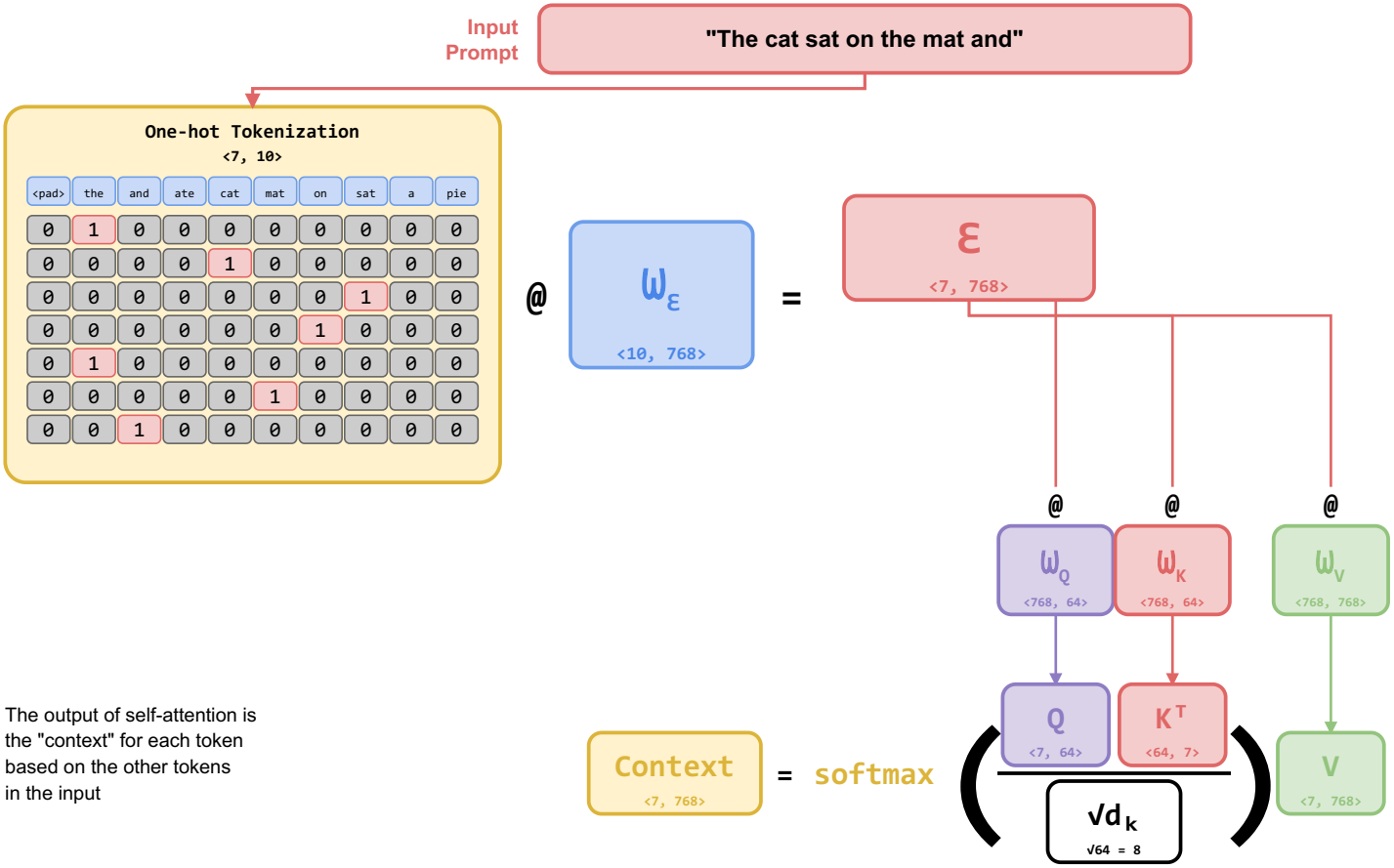
Each row is a measure of how much attention that token should pay to each other token (columns) out of **1**, almost like a "percentage" of your total attention.

Attention during training



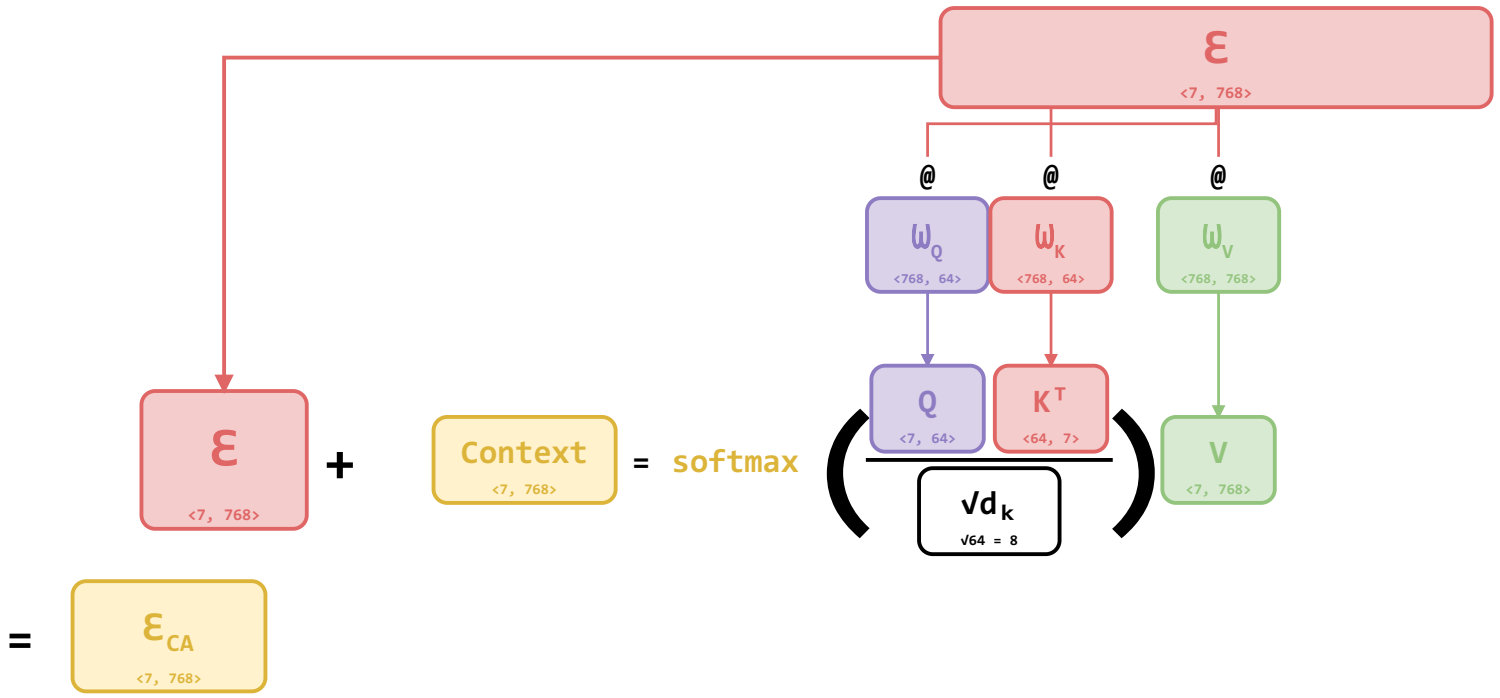
Throughout training, the model will learn to pay attention to different tokens in order to try and minimize the loss by adjusting the weights in W_Q and W_K.

Calculating attention



The output of self-attention is the "context" for each token based on the other tokens in the input

Putting it all together



ε_{CA} = Context-aware embeddings. All of our original embeddings have been "updated" to include the context of all the other tokens in the input.